HW4: JPEG-like Compression with RLE and DCT

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Project Link: https://github.com/RyanCheng98153/video-compression/tree/main/hw4

1. Introduction

In this homework, I implemented **block-based 8×8 DCT compression** with **quantization**, **zigzag scan**, and **run-length encoding (RLE)**. The reconstructed image is obtained through **run-length decoding** and **inverse DCT (IDCT)**.

Two quantization tables (QTable1 and QTable2) are provided, and the effect on **encoded size** and **reconstruction quality (PSNR)** is compared.

This homework demonstrates the **core JPEG compression workflow** and evaluates how different quantization tables influence compression performance.

2. Methodology

2.1 Block-based DCT

The input image (`lena.png`) is divided into (8×8) blocks. For each block, the **Discrete Cosine Transform (DCT)** is computed:

$$F(u,v) = \sum_{x=0}^{7} \sum_{y=0}^{7} \alpha(u)\,\alpha(v)\,f(x,y)\cos\frac{(2x+1)u\pi}{16}\cos\frac{(2y+1)v\pi}{16},$$

where

$$\alpha(u) = \frac{1}{\sqrt{2}} \text{ if } u=0,\ \text{else } 1, \quad \alpha(v) = \frac{1}{\sqrt{2}} \text{ if } v=0,\ \text{else } 1.$$

2.2 Quantization

Each DCT coefficient is divided by its corresponding **quantization table value** and rounded to integers:

$$C_q(u,v) = \mathrm{round}\left(\frac{C(u,v)}{Q(u,v)}\right)$$

This reduces precision and introduces **loss**, enabling higher compression.

2.3 Zigzag Scan & Run-Length Encoding

- **Zigzag scan** converts the 8×8 block into a 1D vector, visiting coefficients from low to high frequency.
- **Run-length encoding (RLE)** compresses sequences of zeros efficiently. Each block is encoded as tuples `(run, value)` until an **End-of-Block (EOB)** marker:

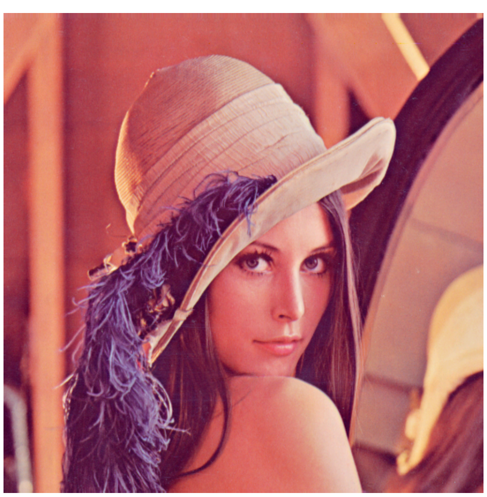
```
Example: [(0,52), (0,3), (5,2), ("EOB",)]
```

2.4 Decoding & Reconstruction

- RLE is decoded to reconstruct the zigzag vector, then converted back to 8×8 blocks.
- **Dequantization** multiplies the coefficients by the original quantization table.
- **Inverse DCT (IDCT)** reconstructs the spatial-domain image.
- Blocks are merged to obtain the final reconstructed image.

3. Results

3.1 Original Image (Lena.png)



3.2 Quantization Tables

- **QTable1**: More fine-grained for low-frequency preservation.
- **QTable2**: Coarser, prioritizing higher compression.

- **Q-Tables**

10	7	6	10	14	24	31	37
7	7	8	11	16	35	36	33
8	8	10	14	24	34	41	34
8	10	13	17	31	52	48	37
11	13	22	34	41	65	62	46
14	21	33	38	49	62	68	55
29	38	47	52	62	73	72	61
43	55	57	59	67	60	62	59

Quantization table 1

10	11	14	28	59	59	59	59
11	13	16	40	59	59	59	59
14	16	34	59	59	59	59	59
28	40	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59

Quantization table 2

3.3 Reconstructed Images

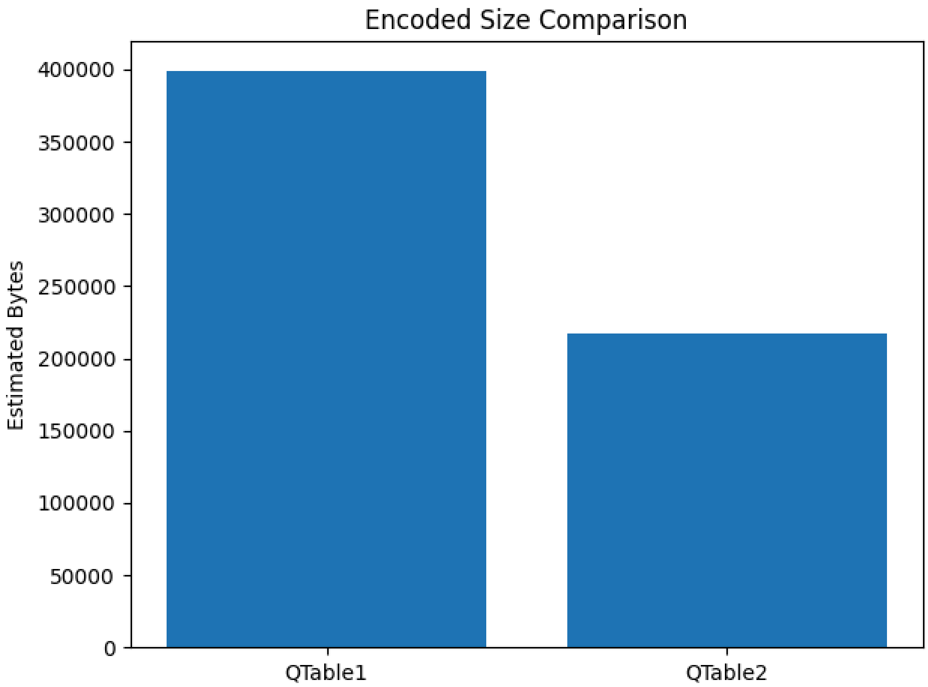
QTable1 Reconstructed Image

QTable2 Reconstructed Image

3.4 Compression Statistics

Quantization Table	Estimated Bytes	PSNR (dB)
QTable1	399,342	35.87
QTable2	216,900	33.38

Bar Plot: Encoded Size Comparison



4. Observations

- **Effect of Quantization Table:**
  - QTable1 retains more low-frequency information → higher PSNR, larger encoded size.
  - QTable2 aggressively quantizes coefficients → smaller encoded size, lower PSNR.
- **Compression Efficiency:**
  Run-length encoding effectively reduces consecutive zeros in high-frequency coefficients. Coarser quantization (QTable2) increases zero sequences, improving RLE efficiency.
- **Image Quality vs Compression:**
  There is a trade-off between compression ratio and visual quality, as indicated by PSNR.
- **Comparison Between QTables:**
  - **Encoded Image Size:** QTable2 achieves ~46% smaller size than QTable1.
  - **PSNR:** QTable1 maintains ~2.5 dB higher reconstruction quality than QTable2.
    This illustrates the trade-off between compression efficiency and visual quality.

5. Conclusion

- Block-based DCT combined with **quantization** and **RLE** provides a simple JPEG-like compression scheme.
- Different quantization tables strongly influence **encoded size** and **reconstruction quality**.
- QTable1 is better for preserving quality, while QTable2 achieves higher compression.
- This homework demonstrates the fundamental steps of **lossy image compression** using DCT and entropy coding.

References

1. JPEG Quantization & DCT: https://www.youtube.com/watch?v=Q2aEzeMDHMA
2. Run-Length Encoding Tutorial: https://q-viper.github.io/2021/05/24/coding-run-length-encoding-in-python/